

Leads: A Deep Learning Approach To Revolutionizing Gas Plant Maintenance With Advanced Anomaly Detection Technology

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Abstract

Pertamina EP Cepu (PEPC) Jambaran Tiung Biru is a major gas producer and a strategic national project in Indonesia. In 2023, the solvent pump, crucial in the selexol solvent regeneration process to reduce H₂S levels, experienced several trips, causing plant shutdowns and production losses. To prevent such failures, PEPC aims to develop an early warning system using anomaly detection to prevent equipment failures and mitigate production losses.

The Leading Equipment Anomaly Detection System (LEADS) was developed using historical solvent pump operating data to model normal conditions. It leverages a Deep Learning LSTM model with an Auto Encoder to reconstruct these conditions and compare them to real-time data. If the delta of Loss MAE between historical and current data exceeds a set threshold, it flags an anomaly. Alerts are sent to the JTB field operations team for proactive measures to prevent failures and shutdowns. LEADS also features plant visualization, historical anomaly trends, operational data mirroring, and automated email notifications.

Using the LSTM model with an autoencoder, the system learns normal operating conditions to establish threshold values for anomaly detection. These anomalies were then clarified and totaled for the number of anomalous results obtained in the four pumps where at P-9027A there were 263 anomalies with a threshold value of 0.108, at P-9027B there were 48 anomalies with a threshold value of 0.128, P-9025A found 3682 anomalies with a threshold value of 0.059, P-9025B found 116 anomalies with a threshold value of 0.145, P-90211A found 12 anomalies with a threshold value of 0.127, and finally at P9011B found 1504 anomalies with 0.103.

The results show a strong correlation between the detected anomalies and actual plant conditions, with each anomaly carefully verified by the operations team. LEADS' ability to reconstruct normal operating conditions in multivariate time series data significantly improves early detection of issues that could lead to solvent pump failures, minimizing the risk of shutdowns and production losses. To further enhance accuracy, it is recommended that specific models be developed for each type of equipment, even if they share the same design and operating parameters. This approach ensures that subtle differences in operation are captured, enabling more precise anomaly detection. The system's flexibility and scalability make it applicable to other critical equipment, offering a robust solution for proactive maintenance and operational reliability.

This paper presents a novel application of the LSTM autoencoder model to detect anomalies in solvent pump operations in gas plants, using multivariate time series data. LEADS reconstructs normal conditions and categorizes anomalies based on context, offering a proactive solution to prevent equipment failures and production losses. This approach enhances reliability in gas processing, particularly for H₂S removal systems, and can be applied to other critical equipment, contributing significantly to operational efficiency and safety.

Introduction

As a subsidiary of PT Pertamina (Persero) dedicated to the exploration and production of natural gas in Cepu, Central Java, the gas extracted from production wells and transported to the JTB plant contains impurities, particularly Hydrogen Sulfide (H₂S), necessitating processing to comply with the standards outlined in the gas sales agreement. The H₂S processing at JTB involves an H₂S removal unit, where the gas is mistreated with a Selexol solvent to reduce H₂S concentrations. This solvent is essential in the process of converting feed gas (raw gas containing H₂S, CO₂, and CH₄) into product gas (processed gas containing CO₂ and CH₄). The solvent facilitates the separation of Carbon Dioxide (CO₂) and Methane (CH₄) from H₂S, ensuring the gas meets the buyer's specifications as stipulated in the Gas Sales Agreement.

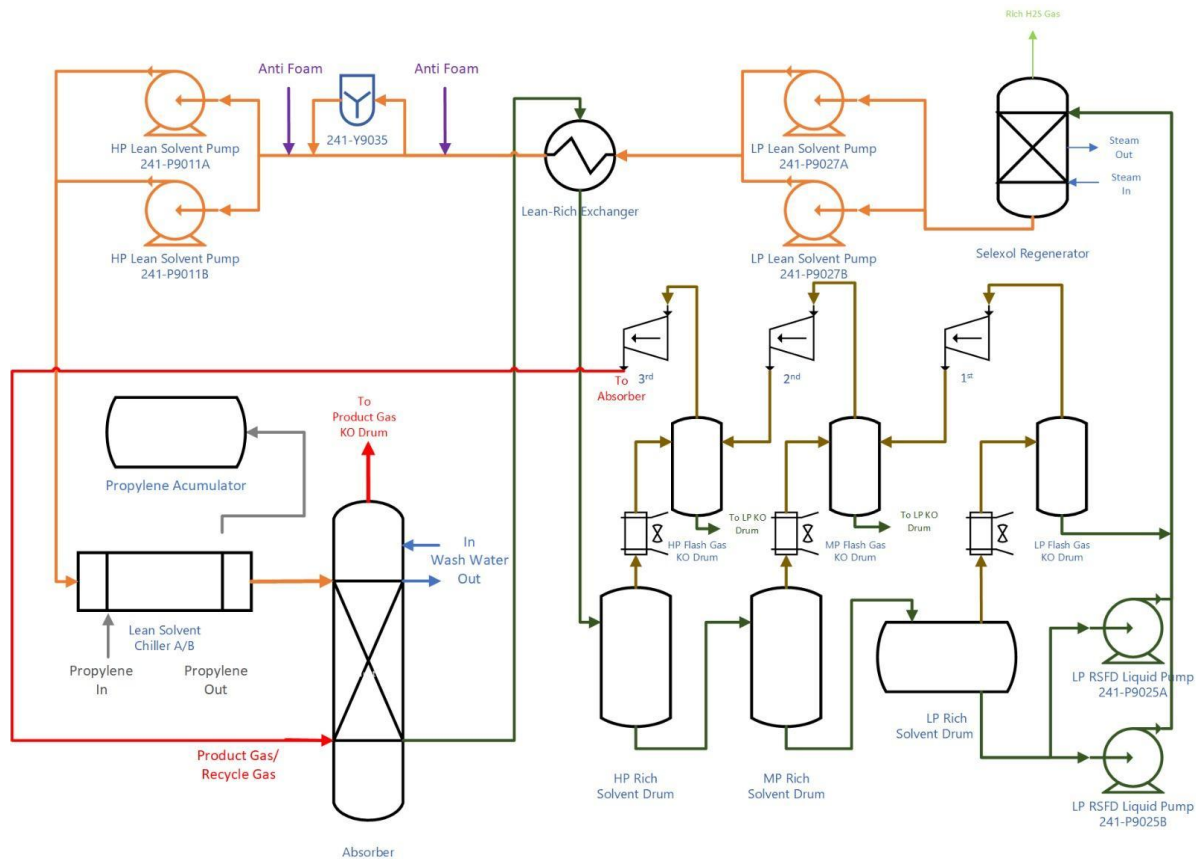


Fig. 1- JTB Unit 241 H2S Removal Selexol Block Diagram.

Fig. 1 depicts the H₂S removal process at JTB, which involves the H₂S removal unit (unit 241). In this unit, gas is treated with Selexol solvent to decrease H₂S concentration. The used solvent is then directed to the regenerator for a stripping process to separate H₂S from the solvent. The circulation of Selexol solvent in the H₂S removal unit is facilitated by three pumps: High Pressure (HP) Lean Solvent pump subunit consists of pumps P9011A and P9011B, Low Pressure (LP) RSFD Liquid Pump subunit consists of pumps P9025A and P9025B, and Low Pressure (LP) Lean Solvent subunit consists of pumps P9027A and P9027B. Each solvent pumps still containing H₂S (Rich Selexol) into the regenerator (T9026) for separation, producing new solvent (Lean Selexol). The regenerated solvent is then pumped by the LP lean solvent pump (P-9027 A/B) to the lean/rich exchanger for heat exchange with the solvent from the absorber. The heated solvent is subsequently pumped by the HP lean solvent pump P-90911 A/B into the lean solvent chiller to prevent overheating before being reprocessed in the absorber. In this cycle, pumps P-9027, P-9025, and P-9011 are critical to the H₂S purification process, and their reliability is essential to avoid operational failures and trips.

The intricate production facilities at the JTB field pose significant challenges, especially regarding the reliability of rotating equipment like pumps. Among these, the solvent pump is crucial for ensuring that the produced gas adheres to the required quality standards. In 2023, the P9027A pump experienced 3 operational trips due to sudden pressure fluctuations, as revealed by inspection performed by the original equipment manufacturer of mechanical seal pumps with a **total ~1.35 million USD financial impact**. Such pressure anomalies in process data can signal impending seal leaks. To address this, machine learning is being applied for anomaly detection in pumps P-9027, P-9025, and P-9011. This approach aims to provide early warnings to prevent trips or unplanned shutdowns, thereby ensuring continuous plant operations.

Anomaly Detection

In this case, ensuring the reliability of critical equipment, such as pumps, is essential for the continuous and efficient operation of industrial plants. This can be achieved by operating the equipment under optimal conditions and implementing rigorous monitoring of process parameters. Anomalies in process data, such as sudden changes in pressure, temperature, or flow rates, can provide early warning signs of potential equipment failures. By leveraging advanced technologies like deep learning, it is possible to develop sophisticated and out of the box anomaly detection systems that can analyze vast amounts of process data in real time.

Anomalies are defined as significant deviations from expected observations, often indicating underlying issues within the system. These deviations can be identified through time-series data, where data points at each time step exhibit unexpected behavior or significantly differ from previous steps (Choi et al., 2021). Anomaly detection in time-series data has been widely studied across various industrial domains, including finance, healthcare, transportation, and energy. For instance, Aljameel et al. (2022) proposed

methods for detecting anomalies in oil and gas pipeline leakage issues using machine learning techniques such as Random Forest, Support Vector Machine (SVM), K-nearest Neighbor, Gradient Boosting, and Decision Tree. Similarly, Sanz-Bobi et al. (2024) utilized Deep Learning Neural Networks (DLNN) to detect anomalies in cooling water pumps of power plants. Apart from that in the study results from Nguyen et al., (2021) about forecasting and anomaly detection approaches using LSTM and LSTM-Autoencoder techniques with applications in supply chain management the results in the LSTM method produced better performance then Deep learning based method which produces in particular, the accuracy, precision, F-scores of LSTM.

These studies highlight the potential of anomaly detection to address real-world problems and enhance decision-making in complex systems, specifically on machine learning methods. For pumps P-9027, P-9025, and P9011, the implementation of anomaly detection involves training models on historical process data to recognize normal operating conditions and identify deviations that may indicate impending failures. This proactive approach allows for early intervention, reducing the risk of unexpected shutdowns and minimizing downtime. Additionally, it enhances the overall performance and reliability of the plant by ensuring that potential issues are addressed before they escalate into critical failures. By integrating these advanced anomaly detection systems into the plant’s monitoring infrastructure such as early warning and data visualization to facilitate further development in a modular manner for facilitate further development with other types of machine learning, at the same time operators can benefit from real-time insights and predictive maintenance capabilities, ultimately leading to improved operational efficiency and reduced maintenance costs.

Where time-series data does not stand independently, it is expected that the most recent time-series data point in the series has an impact on the timestamps of the data points that follow it. In recent years, deep learning has become popular and has been applied in various anomaly detection algorithms (Githinji and Maina, 2023). Thus deep learning can be used to detect anomalies that occur based on existing processes and have a correlation to the failure of a device in this case the 3 pumps in selexol solvent in the H2S removal unit. Therefore, the study was conducted for anomaly detection on the P9025, P-9011, and P-9027 pump objects by implementing LSTM autoencoder for sequential data using the LSTM Encoder-Decoder architecture, as an early warning if anomalies appear in the process data so that trips or unplanned shutdowns can be anticipated and do not affect plant conditions.

Deep Learning Long Short-Term Memory (LSTM) Network

This study proposes the application of deep learning-based methods to tackle the challenges of anomaly detection in complex time-series data, especially when the data lacks sufficient labels. The research employs artificial neural networks, specifically Long Short-Term Memory (LSTM) networks, to model time-series data and identify anomalies by comparing prediction errors against a predefined threshold. LSTM networks, an advanced form of Recurrent Neural Networks (RNN), enhance short-term memory by allowing the use of prior information at specific points. As an extension of RNN, LSTM architecture provides long-term memory capabilities, enabling neural nodes to access all previous information (Wei et al., 2022).

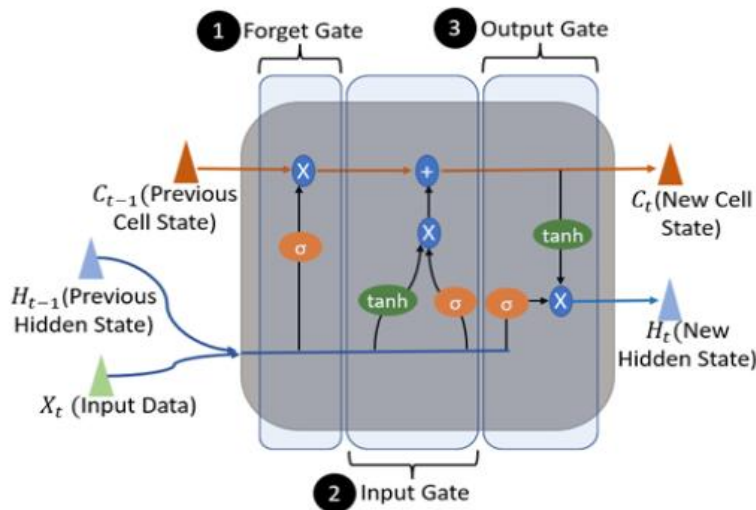


Fig. 2- Fundamental Architecture of a LSTM Network (Wei et al., 2022)

Fig. 2 illustrates the fundamental architecture of an LSTM network, a specialized type of RNN designed to address the vanishing gradient problem that hinders traditional RNNs from learning long-range dependencies in sequential data. The LSTM unit, as illustrated, introduces a “cell state” (C_t), a memory mechanism that persists across time steps, facilitating the flow of information over extended sequences. The network’s operation is governed by three crucial gates: the forget gate (f), the input gate (k), and the output gate (o). These gates, composed of sigmoid neural networks, regulate the flow of information into and out of the cell state. The forget gate determines which information from the previous cell state (C_{t-1}) should be discarded, while the input gate decides which new information from the current input (X_t) should be stored in the cell state. Finally, the output gate controls which information from the cell state should be exposed to the other layers at the current time step.

The mathematical operations within the LSTM unit involve element-wise multiplications (represented by circles with a multiplication sign) and additions. The sigmoid activation function (σ) in the gates outputs values between 0 and 1, representing the extent to which information is allowed to pass through. The hyperbolic tangent activation function ($\tanh$) is used to introduce non-linearity and regulate the values being added to the cell state. The interplay between these gates and activation functions enables the LSTM to selectively retain and utilize information from past time steps, making it particularly effective in tasks involving sequence prediction, natural language processing, and time series analysis. The distinct roles of each gate allow the LSTM to learn complex temporal dependencies and mitigate the issue of vanishing gradients, a common challenge in traditional RNNs, thereby enabling the training of deeper and more powerful networks.

Building upon the foundational understanding of individual LSTM units and their complex gating mechanisms, the focus is shifted to higher-level architectural implementation called the LSTM autoencoder (**Fig. 2**). This network leverages the power of LSTMs to process sequential data uniquely, enabling its application in tasks like anomaly detection. The core idea behind using a reconstruction-based approach for anomaly detection is that normal time series patterns are well-learned and represented in the encoded vector Z , allowing the decoder to reconstruct them accurately. Conversely, anomalous patterns, being different from the norm, are harder to compress into the encoded vector and thus lead to poor reconstruction. The difference between the original input and the reconstructed output is quantified as the “Reconstruction Loss”. A high reconstruction loss for a given input sequence indicates a significant deviation from the learned normal patterns, flagging it as a potential anomaly. A threshold is applied to the reconstruction loss to classify data points or sequences as either “Anomaly” or “Normal”. where the “Anomaly score estimator” calculates this loss for each data point, facilitating the identification of anomalies within the time series.

Methods and Analysis

The Leading Equipment Anomaly Detection System (LEADS) innovation framework employs a structured methodology comprising five key steps, as illustrated in **Fig. 3**. This approach is intended to offer a systematic framework for the development of technology-driven solutions, with a particular focus on machine learning. The LEADS process incorporates principles from the Cross Industry Standard Process for Data Mining (CRISP-DM)^{xx}, which encompasses six primary phases: Business Understanding, Data Understanding, Data Preparation, Modeling, Evaluation, and Deployment.

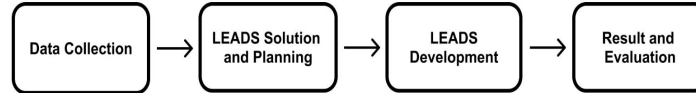


Fig. 3- LEADS Innovation Framework

1. Data Collection

The Data Collection phase involves gathering data from pumps, which serves as input for the machine learning model. Multivariate time series data is collected in real-time at 15-minute intervals daily over the course of one year. This approach aims to capture seasonal patterns in the pump data, aiding the model in learning normal periodic patterns and distinguishing them from anomalous behavior. **Fig. 4** illustrates the data acquisition mechanism, which is indirectly obtained from the Real Time Information System (RTIS) website. This is achieved through data acquisition from the Distributed Control System, Open Platform Communications (OPC) server, and data history server using web scraping methods from web-based historical data that displays real-time updated Human Machine Interface (HMI) data.

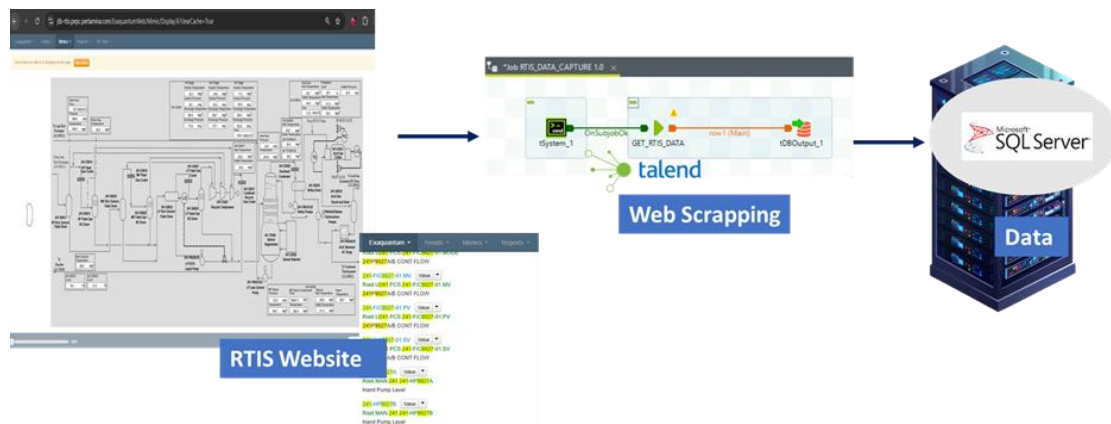


Fig. 4- LEADS Data Acquisition Mechanism

Table 1 outlines the data collection process related to the LP P-9027 A and B solvent pumps. Data was gathered by referencing the Piping and Instrumentation Diagram (P&ID) and subsequently extracting information from the web-based historian data system. Seven parameters, which are presumed to influence the pump process, were collected. These parameters will serve as a reference to identify which ones most significantly contribute to anomalies.

Parameters	Data Type	Unit	Description/Data Preparation
Differential Pressure A/B	Float	Psig	how much water the pump can suck in through its inlet
Suction Pressure A/B	Float	Psig	the force that moves the liquid from the pump to its final destination
Discharge Pressure A/B	Float	Psig	measure the difference between the pressure of the fluid on the inlet side of the strainer and again on the outlet
Flow Rate A/B	Float	gpm	the volume of fluid moved by the pump in one unit of time
Current A/B	Float	Ampere	the amplitude of the motor phase currents
Seal Plan Pressure DE/NDE A/B	Float	Psig	pressure used to generate pressure, maintain pressure, cool and lubricate mechanical seals in
Gas Sales Flow	Float	MMSCFD	a volume of saleable gas that flows through the connected systems point and that can be or has been measured by the metering station

Table 1- LEADS Input Parameters Description

2. LEADS Solution and Planning

The solution and planning phase involves developing solutions and plans to address the identified issues. The solution derived from the Root Cause Analysis includes implementing an alert/early detection system when operational parameters fall between the high and low manufacturer limits. This system serves as a proactive measure to prevent unplanned shutdowns.

To achieve this objective, alternative selection is conducted using multi-criteria decision-making through the Analytical Hierarchy Process (AHP) developed by Thomas Saaty in the 1980s^{xx}. The AHP decision-making process begins with defining the decision problem, developing criteria and alternatives in a hierarchical structure representing the decision problem as shown in **Table 2**.

Table 2- Description of Criteria/Sub-Criteria

LEADS Criteria	Description
1. Life Cycle Cost	Overall operation and maintenance costs while the system is running (per year)
2. Ease of Access and Operation	Ease of access to applications with existing standards, user friendly and with visualizations that are easy to understand
3. Ease of Modification and Replication	Ease of development after implementation and easy to replicate to other pumps, equipment and processes (scalability)
4. Complexity of Development	The level of difficulty of application development includes the use of existing applications
5. Ease of Integration with Existing Systems	Easy connectivity to existing Data Historian and other standard enterprise applications in company

6. Time Development	Time of system development including system setup, engineering, implementation and testing
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Table 2 presents the criteria derived from Focus Group Discussions (FGD) with user experts, IT specialists, and production asset owners. The criteria focus on life cycle costs, ease of access, modification, replication, integration, and development complexity, which are essential for future development with a short development time. Once the selection criteria were defined, a comparison with alternative solutions was conducted. The results in **Table 3** indicated that the internal LEADS solution was the best option, with a higher AHP score (0.463) compared to other alternatives. This choice facilitates further development to maintain the prepared sustainability. The internal development of LEADS was selected because it supports ongoing development to ensure sustainability.

Table 3- Development Priority Ranking

Numbers	Parameter	Weight	Alternatif Solution		
			Leads Internal	Leads External	Non LEADS External
1	Life Cycle Cost	32.94%	Low (Rp 146.045.445)	Medium (Rp 310.500.000)	High (more than USD 92.342)
2	Ease of Access and Operation	16.42%	High	High	High
3	Ease of Modification and Replication	26.16%	High	Medium to High	Medium
4	Complexity of Development	10.78%	Low	Medium	Medium to High
5	Ease of Integration with Existing System	7.90%	High	High	Medium
6	Time Development	5.80%	Low (3 months)	High (9 months)	High (9 months)

After selecting the internal LEADS solution through the AHP method, six planning stages were undertaken to develop LEADS, which are as follows:

a. Requirement Gathering

This phase involves the collection of information, where all necessary data and information required to develop an anomaly detection system for equipment are gathered. This information is obtained through discussions with the development team and relevant experts, including the operations team, engineers, and the maintenance team.

b. Analyze and Design

Following the requirement gathering phase, the next step is analysis and design. In this phase, the collected information is analyzed and selected to identify influential parameters that could potentially cause anomalies. These parameters will be used as time features for the AI model. Additionally, the UI design for the dashboard and the internal system of LEADS will be developed.

Fig. 5 represents the internal system design of LEADS, where the implementation is carried out internally using the standard IT platform of Regional 4 PEPC. This includes Power BI for anomaly data visualization Python as the programming language and environment for AI/ML (LSTM Deep Learning), Talend for data integration from the RTIS and Email alerts, and Relational Database Management System (RDBMS) for the Data Warehouse. Equipment data from the RTIS web will be integrated with Talend, synchronized every 3 minutes to the Data Warehouse within Talend. Talend will then request the pre-built deep learning model to process the data and return the anomaly detection results as a response to Talend's request. Subsequently, these detection results will be forwarded to Power BI for visualization and sent via email alerts. This enables internal company personnel to easily monitor and take action on detected anomalies in the solvent pump.

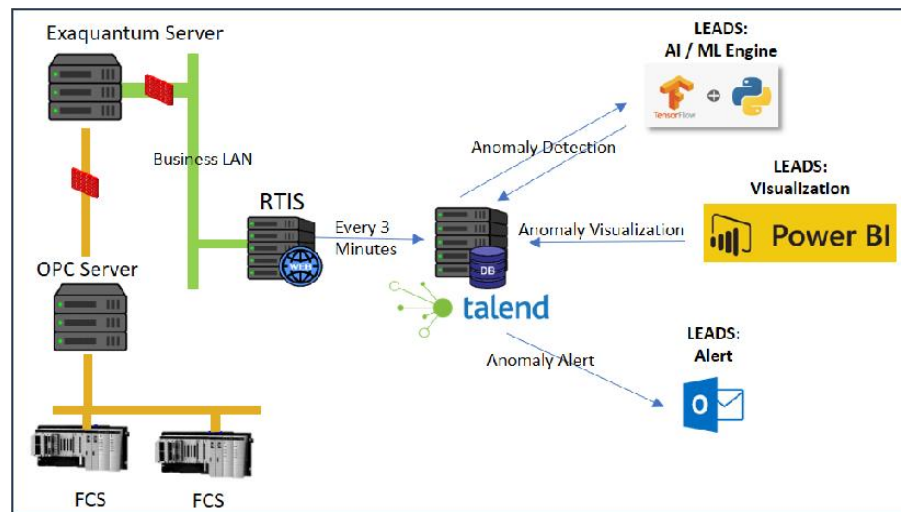


Fig. 5- LEADS Solution Architecture

c. Development of the LEADS Dashboard

In this phase, the LEADS Dashboard is developed using Power BI software. The influential parameters from pumps P-9011, P-9025, and P-9027 A/B as previously detailed in **Table 1** are visualized in the form of charts, graphs, and text to facilitate data retrieval and analysis. The overview display of the LEADS dashboard in Power BI includes information about the six monitored pumps and the P&ID scheme of unit 241, providing an intuitive visualization. Pumps detected with anomalies are marked in red, while those operating normally are marked in green. Additionally, the gas sales parameter, as the final product of the production process, is used as an additional reference for anomaly detection.

d. Develop - Integration of RTIS/Alert

At this stage, we will connect the pump equipment data from the RTIS web to the alert system. As illustrated in **Fig.5**, Talend will retrieve data from the RTIS web and send an alert via Microsoft Outlook to the field operator's email if the machine learning model detects an anomaly. The alert will include details about the anomaly in the pump equipment, such as the date it occurred, the MAE loss parameter value, and the key operating parameters that had an impact. By integrating data from these two systems, we aim to provide faster and more accurate early warnings through the email alert system, thereby supporting more effective decision-making in managing risks and emergency situations.

e. Develop AI Engine

To ensure that the anomaly detection system operates effectively, a model is required that can understand complex patterns in data, particularly in multivariate time series datasets. LEADS has chosen Deep Unsupervised Learning as its approach to analyzing pump equipment data using the Long Short-Term Memory (LSTM) model with an autoencoder.

Python will be the primary programming language due to its flexibility and extensive ecosystem for data processing and developing machine learning models. TensorFlow will serve as the main framework for building and training deep learning models, especially for implementing the LSTM autoencoder architecture, which will be used to detect anomalies based on error reconstruction.

As shown in **Fig.5**, this deep learning algorithm will be integrated with data from the RTIS web. Consequently, any detected anomalies will trigger an alert email, which will include a warning notification and proactive steps to address the issue signaled by the alert.

f. Evaluation

The evaluation phase will assess the quality of the developed system components, including the dashboard, alert system, RTIS integration, and AI engine. This assessment will involve rigorous testing procedures, including functional testing to verify data synchronization, interactivity, and the accuracy of alert notifications under simulated anomaly conditions.

Next, the test will validate the configuration and data transmission between RTIS and SQL server, including manual verification and high-volume load testing to ensure data accuracy and low latency. Finally, the anomaly detection capabilities of the AI engine will be evaluated by assessing the accuracy of anomaly data and threshold values. The model's performance will be optimized through metric evaluation and hyperparameter tuning, using historical data for training and a dedicated anomaly dataset for testing.

In addition to component-specific testing, User Acceptance Testing (UAT) will be conducted to evaluate both the functionality and integration of the system as a whole. This will include testing against real world anomaly events by the JTB operations team to ensure the system's effectiveness in practical scenarios.

3. LEADS Development

The development stage in LEADS is divided into 3 parts, including:

a. Develop - Talend

The LEADS data integration system connects the RTIS to the SQL Server database. This system collects important operational parameters on pump equipment such as flow, pressure, and temperature in real time. This data is crucial for detecting anomalies in critical equipment in the field. When an anomaly is detected, the system automatically sends a warning email to Microsoft Outlook.

As in **Fig. 6**, Talend is a tool for integration between systems and data, Talend provides system service solutions in the first paragraph up to a place to manage data. The Talend data integration tool enables businesses to combine information from various sources efficiently, including databases, flat files, online services, and web APIs.

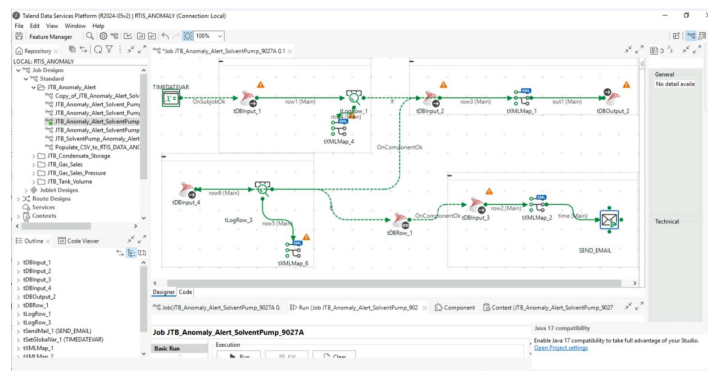


Fig. 6- LEADS System Design Illustration in Talend

Fig. 7 presents a block diagram that elucidates the data acquisition and storage process. The process tag parameters retrieve data from the RTIS web interface and subsequently store this data on the SQL server. This server maintains a dedicated table that mirrors the historical data archived on the RTIS Data Historian server. Data collection occurs at three-minute intervals, providing the basis for anomaly detection within the Talend system.



Fig. 7- LEADS Data Acquisition and Storage Process Block Diagram

Upon the detection of an anomaly, the system, as shown in **Fig. 8** will automatically generate a warning alert via Microsoft Outlook. This alert will serve as an early warning, indicating that operational parameters are approaching predefined threshold limits, thereby enabling proactive intervention before actual thresholds are breached. The alert message will also include recommended actions to mitigate the anomaly. This automated notification system facilitates prompt preventive or corrective measures by the relevant team, minimizing potential damage or operational disruptions to the pump. In essence, the system is engineered to ensure rapid response to abnormal conditions, thereby safeguarding the continuity of the production process and preventing escalation of adverse impacts.

Anomaly Detected!

An anomaly has been detected on LP Lean Solvent Pump P9027A. Below are the details:

Anomaly Details

- **Timestamp:** 2025-01-26 00:45:01
- **Threshold ratio:** 111.5% (Reconstruction loss exceeds threshold by 11.5%)
- **Previous anomaly at:** 2025-01-26 00:36:01

Below are the details on variables that contributed the most* to anomaly and deviation from normal condition:

Variable	Value	Expected Normal Value	Anomaly Deviation	Anomaly Contribution
Suction Pressure (psi) <i>Root U241 PCS 241-P9027-02 PV</i>	39.02	46.03	7.01 (-18.0%)	56.2%
Discharge Pressure (psi) <i>Root U241 PCS 241-P9027-01 PV</i>	288.44	296.47	8.03 (-2.8%)	22.9%
Current (A) <i>Root U241 PCS 241-B9027-01 PV</i>	36.19	35.50	0.69 (+1.9%)	7.4%

* The most dominant cause: 80% of total anomaly contribution.

Visual Summary

Please check the attached image file for the visual of anomaly.

[See more details on dashboard](#)

This is an automated alert. Please do not reply to this email.
Please use company's network or VPN connection to access the dashboard.
If you have any questions, contact the support team at IT.Dama.Regional@searamina.com

Fig. 8- LEADS Alert Notification Message

b. Develop - Dashboard

The LEADS system incorporates a Power BI-developed dashboard to visualize the data processed by Talend. This dashboard presents two distinct views, as illustrated in **Fig. 9**. The first view provides a comprehensive overview of the unit 241 H2S removal selexol system process, including a schematic representation of the solvent pumps (P-9027 A/B, P-9011 A/B, and P-9025 A/B) under monitoring. The second view focuses on historical anomaly detection. It displays a chart incorporating data from seven process parameters and historical anomaly records stored in the SQL Server database.

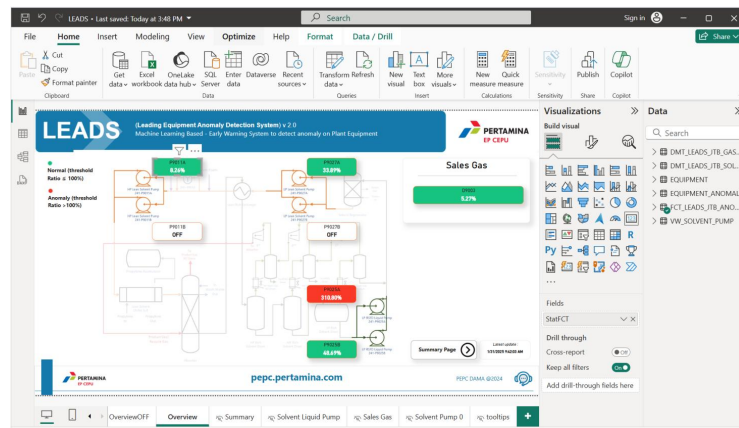


Fig. 9- LEADS Power BI Dashboard View

The trend of each pump's operational parameters can be ascertained from the time-series graphs presented in **Fig.10**. This data visualization enables continuous monitoring of parameter value fluctuations, thereby facilitating the identification of anomaly occurrence times for each operational parameter.

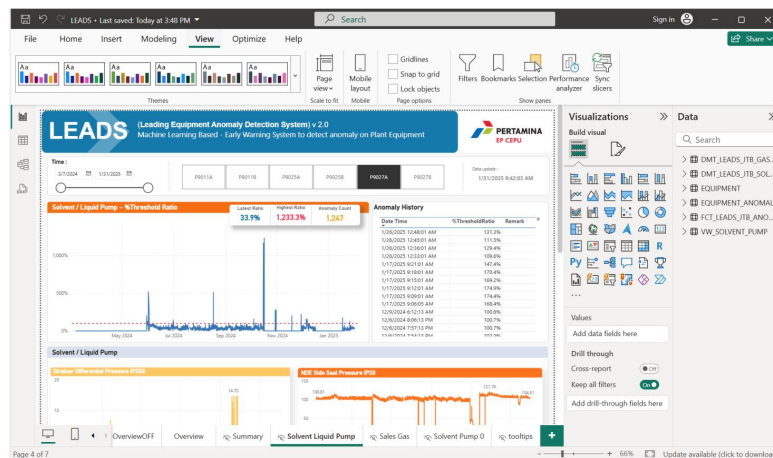


Fig. 10- LEADS Detail Pump Operational Parameter View

c. Develop ML/AI Engine

The stage of developing a machine learning model will be divided into several parts using the CRISP-DM method; each part has its function to make the model detect accurately as depicted in **Fig. 10**. The model for LEADS anomaly detection uses a deep learning model with a Long short-term memory (LSTM) algorithm with an autoencoder, where this model is expected to be able to study data in depth in the form of a multivariate time series and detect anomalies in each data. The development of a machine learning model will be divided into several stages using the CRISP-DM methodology. Each stage serves a specific purpose to enhance the model's accuracy, as illustrated in **Fig.11** The model designed for LEADS anomaly detection utilizes a deep learning approach, specifically a Long Short-Term Memory (LSTM) algorithm combined with an autoencoder. This model aims to analyze multivariate time series data comprehensively and effectively identify anomalies within that data.

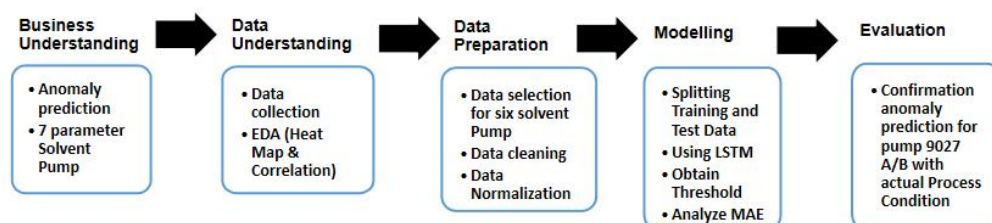


Fig. 11- LEADS Steps to develop ML/AI Engine

Business Understanding is the stage of data collection and setting goals of the data mining process.

During the **Data Understanding** phase, we will separate the operating parameters of the pump, currently combined in **Fig. 13**, into seven individual features (x). Alongside this, we will include the datetime format to indicate when each event or observation occurs (y). After this separation, we will conduct Exploratory Data Analysis (EDA) to gain a better understanding of the data's characteristics and correlations.

Data Preparation involves selecting, cleansing, and normalizing data that has already been explored. It is essential to format the data according to specific requirements. At this stage, historical data of seven operating parameters on the pump as in **Table.1** will be taken from the RTIS Website in the form of a csv file with the aim of becoming an input model to predict anomalies. The data especially for the pump 9027 A/B, we will ensure that there are no empty entries, and we will normalize the data on a scale from -1 to 1. We will apply data normalization techniques to eliminate the impact of different scales on the readings of each feature in the multivariate time series. This process will reduce the execution time and computational complexity during the training of deep learning models. The formula for data normalization is as follows:

$$Z_i = \frac{X_i - \mu}{S} \dots\dots\dots(\text{Eq.1})$$

as described in **Eq.1** Z_i is the normalized value in the range [-1, 1]. X_i represents the actual data point, μ denotes the mean value, and S indicates the standard deviation.

The **Modeling Phase** of the Data Pump, following the data preparation stage, involves dividing the dataset into training and test sets. We allocate 60% of the total data for training and the remaining 40% for testing. Next, the training process begins, utilizing a deep learning algorithm known as LSTM-AE (Long Short-Term Memory Autoencoder) with a specific autoencoder model architecture, as illustrated in **Fig.13** below.

Layer (type)	Output Shape	param #
reshape_input (Reshape)	(None, 1, 9)	0
lstm_encoder_0 (LSTM)	(None, 1, 16)	1,664
lstm_encoder_1 (LSTM)	(None, 4)	336
Repeat Vector	(None, 1, 4)	0
lstm_decoder_1 (LSTM)	(None, 1, 4)	144
lstm_decoder_0 (LSTM)	(None, 1, 16)	1,344
time_distributed (Time Distributed)	(None, 1, 9)	153
reshape_output (Reshape)	(None, 9)	0

Fig.12 Layers of LSTM-AE Models

In this model, LSTM serves as both the encoder and decoder. The encoder component employs LSTM to process and compress sequential data into a more compact representation, referred to as the latent space. This is achieved using two layers with 16 and 9 neurons, respectively. The decoder, meanwhile, uses LSTM to reconstruct the original data from this compressed representation through two layers consisting of 9 and 16 neurons. Additionally, we introduce a Repeat Vector layer (Layer 3) to replicate the 1×16 vector for each timestep. Given that our model has a timestep size of 9, Layer 2 generates four copies of the encoded features, forming two-dimensional vectors with dimensions of 9×16 .

Each encoded feature is processed by an individual LSTM cell. The outputs from all LSTM cells are then combined into a one-dimensional vector in the Time Distributed Dense Layer, which generates the final output. To evaluate how well the LSTM-AE model reconstructs the encoder, we calculate the difference between the test data and the reconstructed data, known as the reconstruction error. We use the Mean Absolute Error (MAE) algorithm, represented by the following equation:

$$Loss (MAE) = \frac{\sum_{i=1}^n |x_i - \hat{x}|}{n} \dots\dots\dots(\text{Eq.2})$$

as described in **Eq.2** where n is the total number of samples, x_i represents the original input given to the encoder, and $\hat{x}$ denotes the output produced by the decoder. The Mean Absolute Error (MAE) will be used to determine the optimal threshold based on the reconstruction error rate evaluated for each data point in multivariate time series sequences. This process aims to identify anomalies.

The threshold calculation is based on the assumption that the MAE values for reconstruction follow a normal distribution. As illustrated in **Eq.3**, the threshold is calculated using the percentile of the normal distribution. A probability of 99% is applied to establish the upper limit of errors considered normal. If the reconstruction error of a data point exceeds this predetermined threshold, that data point is classified as an anomaly.

$$Threshold = \mu(Loss\ MAE) + z * \sigma (Loss\ MAE) \dots\dots\dots(Eq.3)$$

where μ represents the average reconstruction loss (MAE), z is the z-score corresponding to a 99% confidence level, and σ denotes the standard deviation of the reconstruction loss (MAE).

Evaluation is the final stage in the data mining process. At this stage, an evaluation will be carried out in the form of fine tuning on the model that has been created, namely the model will be tried to be trained using other datasets containing anomalies and verifying the results of anomaly detection on the model with actual field events, this process requires a field operator as a verifier of the anomaly detection results.

Results

Fig.13 presents the front page of the Leading Equipment Anomaly Detection System (LEADS) dashboard, designed to display real-time conditions related to anomalies in solvent pumps and gas sales. Six pumps were monitored for anomalies: P9027A, P9027B, P9025A, P9025B, P9011A, and P9011B. Components that are operational and have a threshold of less than 100% are displayed in green, indicating normal functioning. In contrast, components with a threshold ratio exceeding 100% are marked in red, as illustrated in Figure P9025A.

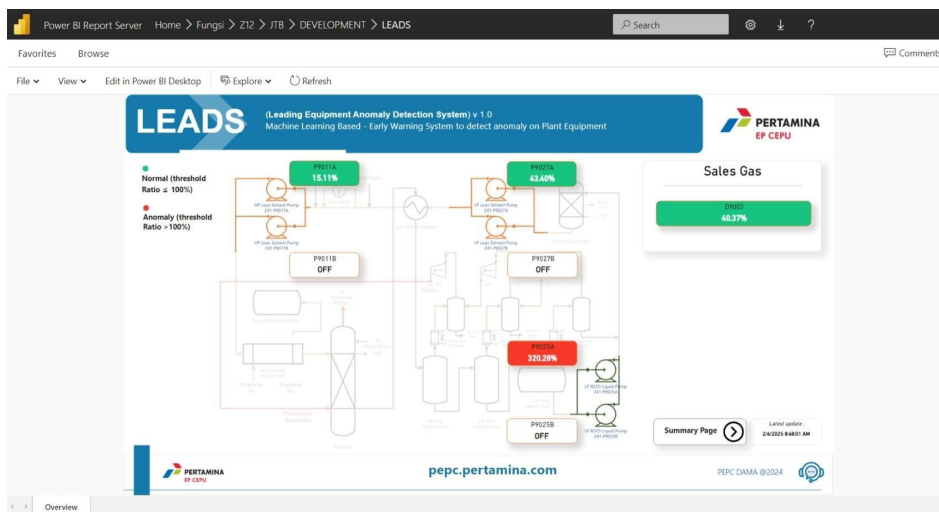


Fig. 13-Leading Equipment Anomaly Detection System (LEADS) Dashboard

Fig.14 provides a detailed view of one of the pumps. It includes a graph that displays the threshold ratio of the solvent or liquid pump, along with key information such as the latest ratio, the highest ratio, and the number of anomalies detected. This page also presents a detailed graph of each parameter and historical data on past anomalies that have occurred with the pump, making it easier for operators to verify any issues that arise.

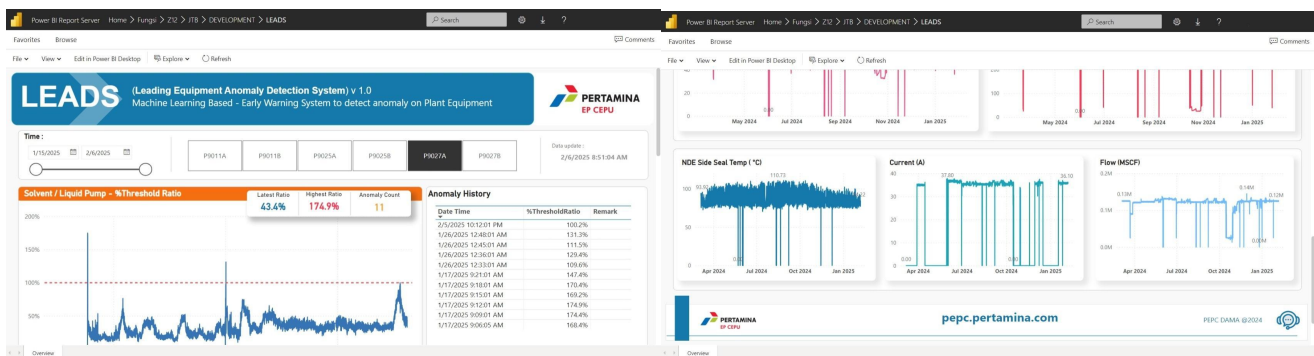


Fig 14-Detail page of P9027A pump

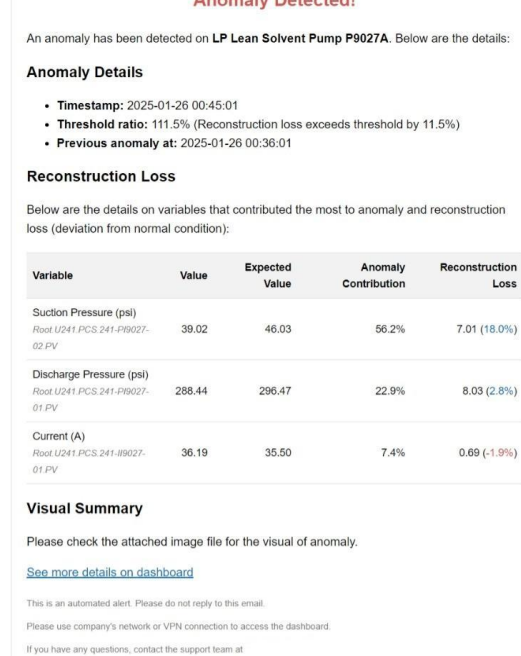


Fig.17 shows the comparison results of training loss (MAE) and validation loss (MAE) when training the model using pump data from one type of pump, namely the P9027 pump. This graph shows that the MAE loss and validation MAE loss tend to be stable and both have a close distance. This indicates that the model performance is quite accurate.

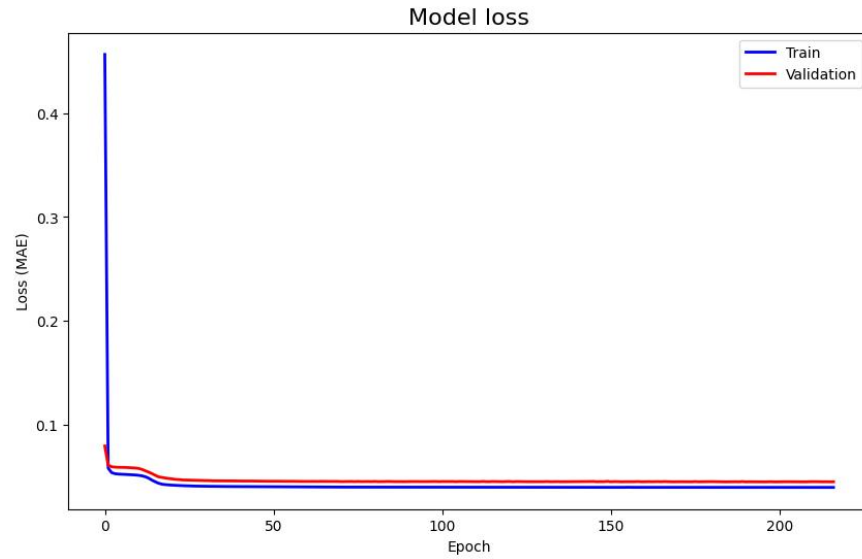


Fig 17-Comparison between Training Loss (MAE) and Validation Loss (MAE) on P9027A/B pump

This is proven by the results of reconstruction loss on a logarithmic scale, as shown in **Fig.19**. Namely, the distribution of loss resembles a normal distribution, so it can be said that the model is good enough.

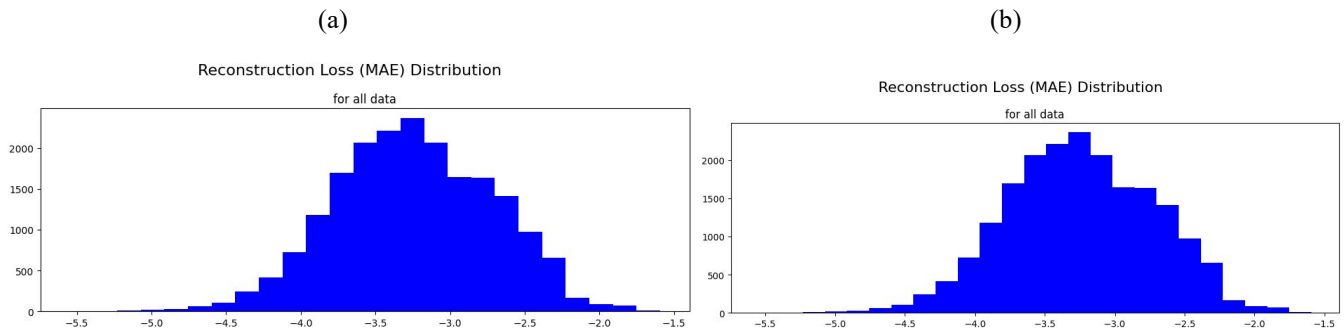


Fig 18-Reconstruction Loss (MAE) distribution results for train data and test data on P9027A (a) and P9027B (b)

The following plot illustrates the reconstruction loss on a logarithmic scale, along with the anomalies detected at the P907A pump from January 2024 to January 2025, with data collected at 15-minute intervals, as shown in **Fig.19**. During this period, there are 264 anomalies highlighted in red. These anomalies are identified when the reconstruction loss value exceeds the threshold of 0.10853501191977315.

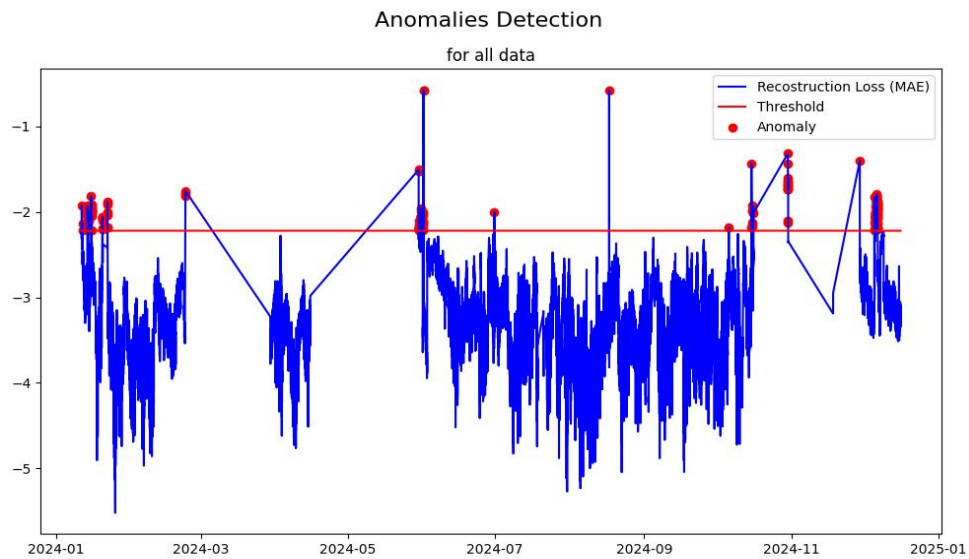


Fig 19-Anomaly Detection Results on P9027A Pump

In the following **Fig.20** below are the top five sample data from 264 anomalies detected on the P9027A pump, along with the values of the seven parameters that affect the solvent pump.

	differential_pressure	seal_pressure	suction_pressure	discharge_pressure	temperature	current	flow_mscf
datetime							
2024-01-11 21:30:00	0.817228	106.9353	39.17393	283.1572	82.01967	36.47661	126316.00
2024-01-12 04:00:00	1.521557	107.6472	42.24725	313.0451	77.75644	32.02934	95536.59
2024-01-12 04:15:00	1.605658	107.6472	43.16098	316.9532	78.26395	32.31812	94252.40
2024-01-14 06:00:00	1.454960	105.5024	47.86073	303.3532	81.16428	33.72355	111106.50
2024-01-14 06:15:00	1.465814	105.2475	47.77683	302.6447	81.16428	33.79622	112173.80

Fig 20-Top Five Data Anomalies Detected on P9027A Pump

The following plot illustrates the reconstruction loss on a logarithmic scale, along with the anomalies detected at the P907A pump from January 2024 to January 2025, with data collected at 15-minute intervals, as shown in **Fig.21**. During this period, there are 264 anomalies highlighted in red. These anomalies are identified when the reconstruction loss value exceeds the threshold of 0.10853501191977315.

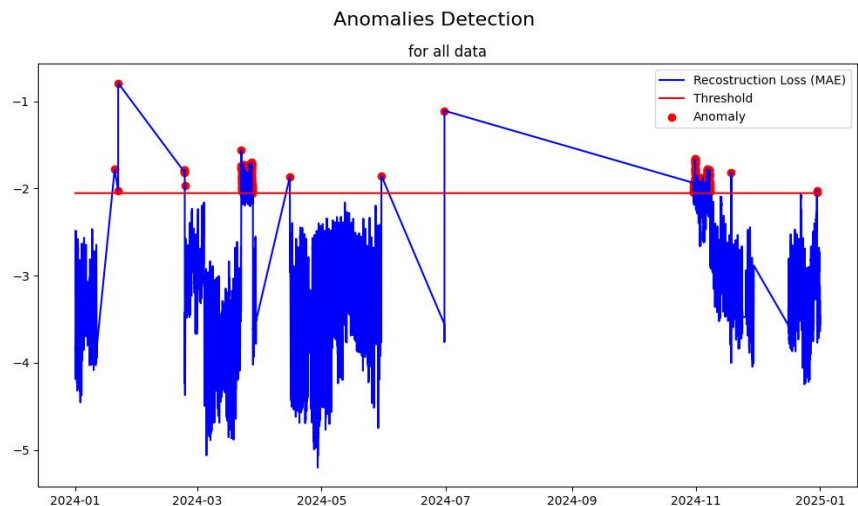


Fig 21-Anomaly Detection Results on P9027B Pump

In the following **Fig.22** below are the top five samples from 48 anomalies detected on the P9027B pump are presented, along with the values of seven parameters that affect the solvent pump.

	differential_pressure	seal_pressure	suction_pressure	discharge_pressure	temperature	current	flow_mscf
datetime							
2024-01-22 08:30:00	0.309420	105.7363	33.40611	316.55860	82.78477	30.94645	90209.94
2024-01-22 08:45:00	14.664680	106.7486	32.66357	24.91474	83.63132	30.93691	25740.28
2024-02-24 00:45:00	0.207562	106.5228	40.38760	320.00710	81.51542	27.90470	72194.94
2024-02-24 01:00:00	0.207562	106.7936	40.01017	319.32040	81.76649	27.85778	71962.68
2024-02-24 01:15:00	0.222408	107.0628	39.83532	318.71840	82.01152	27.71406	71819.69

Fig 22-top five data anomalies detected on P9027B pump

In **Table 4** below, we present the results of anomalies detected in each solvent pump, along with the corresponding thresholds for each pump. From this table, it is evident that the P9025A pump had the highest number of anomalies, totaling 3,682.

Table 4-Number of Anomalies on Each Solvent Pump

Pump	Anomaly Count	Data Count	thresold
P9027A	263	19855	0.10853501191977315
P9027B	48	12987	0.12828787755746715
P9025A	3682	20610	0.05988156707432593
P9025B	116	12566	0.14449655958741475
P9011A	12	8052	0.127472213004012
P9011B	1504	2688	0.10324008718540229

The detected anomalies will be confirmed through operational evaluation. This process will analyze the data from the seven solvent pump parameters, comparing it with the monitoring logbook and historical pump data corresponding to the time of the anomaly, as illustrated in **Table 5**. The operational confirmation results indicate that on January 11, 2024, at 21:20, a Start-Up occurred on the P9027A pump. At 21:45, the pump tripped due to a high differential pressure (DP) reading on the suction strainer. The following day, at 03:16, the selexol flow rate was gradually increased. These findings demonstrate that the detected anomaly successfully identifies ongoing issues with the solvent pump at that time.

Table 5 - Sample confirmation results with the operator regarding the anomaly that occurred in pump P9027A.

Datetime	Differential Pressure	Seal Pressure	Suction Pressure	Discharge Pressure	Temperature	Flow	Current	Reason/ Note
2024-01-11 21:30:00	817.228	1.069.353	3.917.393	2.831.572	8.201.967	3.647.661	3.647.661	21:20 Start 241P9027A LP lean Solvent 21:45 241P9027A trip, low current due to High DP Suction strainer
2024-01-12 4:00:00	1.521.557	1.076.472	4.224.725	3.130.451	7.775.644	3.202.934	3.202.934	03.16 Start P9027A, gradually increase the flow rate of selexol
2024-01-12 4:15:00	1.605.658	1.076.472	4.316.098	3.169.532	7.826.395	3.231.812	3.231.812	gradually increase the flow rate of selexol

2024-01-14 6:00:00	145.496	1.055.024	4.786.073	3.033.532	8.116.428	3.372.355	111106.5	gradually increase the flow rate of selexol from 92 into 95 kbpd
2024-01-14 6:15:00	1.465.814	1.052.475	4.777.683	3.026.447	8.116.428	3.379.622	112173.8	gradually increase the flow rate of selexol from 92 into 95 kbpd
2024-01-14 6:30:00	1.462.183	1.052.475	4.781.453	3.027.397	8.066.933	3.381.672	110988.4	gradually increase the flow rate of selexol from 92 into 95 kbpd

Leading Equipment Anomaly and Detection System (LEADS) has also been subjected to quality assurance and quality control to ensure that the entire system is running as desired and has good performance. Some of the quality assurances that have been carried out are as follows:

1. Ensuring that data sent from RTIS is successfully received in SQL Server without any errors with a delivery latency of less than 2 seconds.
2. Ensuring the stability of the LSTM model used and ensuring that it can detect anomalies with verification results confirmed by the operator as many as **184 of the 263 anomalies** detected on the P9027A pump and **211 of the 211 anomalies** on the P9027B pump.
3. Ensuring that the data displayed along with all interactive elements displayed on the dashboard are correct and functioning properly
4. Ensuring that email alerts can be sent correctly after anomaly detection

Discussion

Based on the results presented above, the Leading Equipment Anomaly Detection System (LEADS) has been able to provide a solution to the problems that exist in solvent pumps in the JTB Field by being able to detect anomalies in solvent pumps using the LSTM Autoencoder model which can detect 263 anomalies in the P9027A pump with a threshold of 0.10853501191977315, 48 anomalies in the P9027B pump with a threshold of 0.12828787755746715, **12 anomalies in the P9011A pump** with a threshold of 0.127472213004012, **1504 anomalies in the P9011B pump** with a threshold of 0.10324008718540229, **3862 anomalies in the P9025A pump** with a threshold of 0.10324008718540229, and **3862 anomalies in the P9025A pump** with a threshold of 0.10853501191977315. 0.05988156707432593, and **116 anomalies on the P9025B pump** with a threshold of 0.14449655958741475. The anomalies that have been detected have been confirmed by the operator, as many as **184 of the 263 equipment anomalies on the P9027A pump** and **211 of the 211 equipment anomalies on the P9027B pump**. The results of the confirmation of the anomalies by the operation prove that the detected anomalies are able to indicate something is happening to the solvent pump. This LEADS system is also able to provide warning notifications to the stakeholders when an anomaly is detected on the solvent pump.

The Leading Equipment Anomaly Detection System (LEADS) provides several benefits to the company. It can save unscheduled maintenance and internal development costs amounting to up to USD 476,157. Additionally, it helps reduce high-level risks associated with pump repairs due to leaks in the mechanical seals, which can potentially lead to process fluid leaks into the environment. In 2023, JTB Field experienced two mechanical seal leaks at 241-P9027, which contributed to actual plant availability of 93.07%, with a reduction of 0.16%. By utilizing Long Short-Term Memory (LSTM) with an Autoencoder to detect anomalies through Mean Absolute Error (MAE) and standard deviation thresholds. This approach has successfully prevented mechanical seal failures in the solvent pumps at 241-P9027, resulting in zero lost production hours (LPO). Furthermore, this implementation has significantly boosted the confidence and morale of workers, increasing it by 97% among those who initially had low awareness of the Key Performance Indicator (KPI) PTK-041 and the operation of solvent pumps to prevent unplanned shutdowns.

There is an opportunity to develop the Leading Equipment Anomaly Detection System (LEADS) feature, specifically focusing on the initial classification of previously detected anomalies. This classification will categorize anomalies into several types, including Start-up, Nomination Changes, Process Changes, and Equipment Anomalies. The classification is based on information provided by operators, which will then be matched with the predetermined categories. We currently utilize a classification method that involves supervised machine learning, specifically using Decision Trees and Logistic Regression. To enhance scalability and adaptability, LEADS is envisioned as a highly scalable modular system solution, allowing for seamless integration with evolving machine learning models and operational requirements.

Additionally, We are currently developing an advanced rule-based system to evaluate whether the pump is functioning as intended. This evaluation is based on pump specification data, which includes minimum flow rates and the specifications of the impeller used in the solvent pump. The pump specifications can be translated into several analytical rules. These rules may involve assessing the current condition of the pump's head about its flow, determining if the pump is experiencing cavitation, and checking whether the

pump's performance aligns with the specifications of the impeller used. Additionally, many other rules merit further exploration.

The classification results and established rules are anticipated to serve as proactive solutions within prescriptive analytics systems. This methodology encourages proactive actions, such as generating work orders based on prescriptive analytics findings. Furthermore, the development of this analytical method will not end at this stage; it will continue to evolve to maximize the benefits that organizations can achieve.

Conclusion

LEADS were developed as a breakthrough solution using the LSTM autoencoder model to detect anomalies in solvent pump operations in gas plants, using multivariate time series data and modular application systems. LEADS reconstructs normal conditions and categorizes anomalies based on context, offering a proactive solution to prevent equipment failures and production losses. This approach enhances reliability in gas processing, particularly for H₂S removal systems, and can be applied to other critical equipment, contributing significantly to operational efficiency and safety.

The implementation of the Leading Equipment Anomaly Detection System (LEADS) has demonstrated significant effectiveness in detecting anomalies in solvent pumps within the JTB Field. By utilizing the LSTM Autoencoder model, the system successfully identified anomalies across multiple solvent pumps, with the detected anomalies validated by operators. This confirmation underscores the reliability of the system in identifying operational issues and enabling timely intervention. Furthermore, LEADS provides critical warning notifications to stakeholders, enhancing the overall responsiveness to potential equipment failures. Beyond anomaly detection, LEADS also offers substantial cost savings in the financial sector and given operational benefits by mitigating high-level risks associated with mechanical seal failures, preventing environmental hazards and enhancing plant availability. The implementation has also fostered greater worker confidence regarding awareness of key performance indicators and equipment operation.

The potential for further development of LEADS is significant. Future enhancements include the classification of detected anomalies into categories such as Start-up, Nomination Changes, Process Changes, and Equipment Anomalies. This classification will leverage operator feedback to refine the categorization process, ensuring accurate anomaly identification and response. Additionally, the introduction of a rule-based system enables a comprehensive assessment of pump performance, incorporating key specifications such as flow rates, impeller characteristics, and cavitation analysis. These analytical rules will further strengthen the system's capability to assess equipment conditions and proactively generate work orders for maintenance.

In conclusion, LEADS has proven to be an invaluable tool for predictive maintenance, operational efficiency, and risk mitigation in solvent pump management. Continuous advancements in anomaly classification and rule-based analysis will further enhance its capabilities, ensuring sustained operational improvements and long-term benefits for the organization.

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